A COVER OF THE REGULAR TETRAHEDRON BY 640 CYLINDERS BELOW BANG'S HALF-AREA BOUND

[AUTHOR NAME]

Abstract. OpenAI's preprint *Finite angular cylinder covers below the half-area bound* (`openai/math`, family 100) disproves Bang's half-area question: a regular tetrahedron K admits a finite cover by cylinders whose perpendicular bases have total area below $\frac{1}{2}A_{\min}(K)$. Its explicit threshold $\varepsilon \le 1/2000$ yields $2\lceil 2/\varepsilon^2\rceil = 16{,}000{,}000$ cylinders. Keeping the same angular construction and the same (parameter-free) sector-selection lemma, but replacing the global radial estimate by per-sector cutoffs certified by 102,400 exact rational linear programs, we obtain a cover by 640 cylinders ($\varepsilon = 2/5$) with $\sum |B_i|/A_{\min}(K) < 0.4999773399$.

1. Setting

A cylinder is $C = B + \mathbb{R}u$ with u a unit vector and $B \subset u^\perp$ measurable; $|B|$ is its (perpendicular) base area. For a convex body $K \subset \mathbb{R}^3$, $A_{\min}(K)$ is the minimum area of an orthogonal projection of K onto a plane. Bang [1] asked, as recorded by Bezdek and Litvak [3, Introduction] and Bezdek [2, Problem 3.1], whether every finite cylinder cover satisfies $\sum_i |B_i| \ge \frac{1}{2}A_{\min}(K)$; the two-cylinder cover of a regular tetrahedron attains equality. The family-100 preprint [4] answers this negatively. Related work. An independent parameter-window optimisation within the original certificate gives 10^6 cylinders [5]: keeping the preprint's coverage condition $\eta \ge (2\tau+\tau^2/4)/(2(1-\tau))$ and area condition $\eta < 1/480 - \tau^2$, the choice $\tau = 1/500$, $\eta = 1/498$, $2/n = \tau^2$ gives $2n = 10^6$ cylinders (a factor 16 fewer), with exact coverage budget $1751/249000000$ and area saving at least $2959/5187500000000$ of $A_{\min}(K)$; we thank its authors. Our improvement to 640 replaces the coverage argument itself (per-sector truncations T_j and exact pairwise LPs over 320^2 sector pairs), which allows $\varepsilon = 2/5$ instead of $\varepsilon \le 1/2000$; any choice inside the original certificate's window still needs $\tau \lesssim 1/482$, i.e. more than $9 \cdot 10^5$ cylinders.

We use its coordinates: $H = \sqrt{2}$,

$$K = \{(x, y, Ht) : 0 \le t \le 1,\ |x| \le 1 - t,\ |y| \le t\},$$

a regular tetrahedron of edge 2 with $A_{\min}(K) = H$ [4, §2, `lem:projection-minimum`]. Fix $\varepsilon > 0$, an integer n, $q_j = -1 + 2j/n$ $(0 \le j \le n)$,

$$\alpha_j = \tfrac{1}{4}(1 + q_j q_{j+1}), \qquad \beta_j = \tfrac{1}{4}(q_j + q_{j+1}),$$

and *arbitrary* positive cutoffs $T_0, \dots, T_{n-1}$. The $2n$ cylinders are $P_j + \mathbb{R}v_j$ and $Q_j + \mathbb{R}w_j$ with $v_j = (1, \varepsilon\alpha_j, H\varepsilon\beta_j)$, $w_j = (-\varepsilon\alpha_j, 1, H\varepsilon\beta_j)$ and intercept triangles

$$P_j = \{(0, y_0, Ht_0) : 0 \le t_0 \le T_j,\ q_j t_0 \le y_0 \le q_{j+1} t_0\},$$
$$Q_j = \{(x_0, 0, H(1 - s_0)) : 0 \le s_0 \le T_j,\ q_j s_0 \le x_0 \le q_{j+1} s_0\}.$$

The preprint takes $T_j = \frac{1}{2} + \varepsilon^2(\eta + \max_{[q_j, q_{j+1}]} d)$ with $d(q) = q^2(1+q^2)/16$, $\eta = 10^{-3}$, $n = \lceil 2/\varepsilon^2 \rceil$, and proves coverage for $\varepsilon \le \eta/2$ [4, §3, `prop:coverage`].

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2. Two parameter-free facts

Lemma 1 (membership)**.** *$p = (x, y, Ht)$ lies in $P_j + \mathbb{R}v_j$ iff, with $t_0 = t - \varepsilon x\beta_j$ and $y_0 = y - \varepsilon x\alpha_j$, $0 \le t_0 \le T_j$ and $q_j t_0 \le y_0 \le q_{j+1} t_0$. It lies in $Q_k + \mathbb{R}w_k$ iff, with $s_0 = 1 - t + \varepsilon y\beta_k$, $x_0 = x + \varepsilon y\alpha_k$, $0 \le s_0 \le T_k$ and $q_k s_0 \le x_0 \le q_{k+1} s_0$.*

Proof. The first coordinate of v_j is 1, so $p - \lambda v_j$ meets the plane $\{x = 0\}$ exactly for $\lambda = x$, at $(0, y_0, Ht_0)$. The second family is symmetric. □

Write W_j (resp. V_k) for the set of $p \in K$ satisfying the two *angular* inequalities of Lemma 1 for sector j (resp. k).

Lemma 2 (first crossing; [4, proof of `prop:coverage`, lines 13–45 of `coverage.tex`])**.** *For every $\varepsilon > 0$ and $n \ge 1$, $K = \bigcup_j W_j = \bigcup_k V_k$, and $p \in W_j$ implies $t_0 \ge 0$, $p \in V_k$ implies $s_0 \ge 0$.*

Proof. Since $\alpha_j - q\beta_j = (1 - q^2)/4$ for $q \in \{q_j, q_{j+1}\}$, the inequality $q_i t_0 \le y_0$ ($i \in \{j, j+1\}$) is equivalent to $F_i \le y$ with $F_i = q_i t + \varepsilon x(1 - q_i^2)/4$, independent of j. As $F_0 = -t \le y \le t = F_n$, the first $i \ge 1$ with $F_i \ge y$ gives $F_{i-1} \le y \le F_i$, i.e. $p \in W_{i-1}$; no monotonicity of (F_i) is used. Subtracting the two inequalities gives $(q_{j+1} - q_j)t_0 \ge 0$. The second family is identical with $G_i = q_i(1 - t) - \varepsilon y(1 - q_i^2)/4$. □

Neither lemma uses smallness of ε: the only properties used are the exact identity above and $\phi(\pm 1) = 0$ for $\phi(q) = (1 - q^2)/4$.

3. The reduction

For a pair (j, k) let

$$Z_{jk} = \max\{\min(t_0 - T_j,\ s_0 - T_k) :\ p \in W_j \cap V_k\},$$

a linear program in (x, y, t, z) with 12 constraints (six for K, two wedge inequalities per family, two for z); $W_j \cap V_k \neq \emptyset$ always, and the maximum exists since K is compact.

Theorem 3. *If $Z_{jk} < 0$ for all $0 \le j, k < n$, the $2n$ cylinders cover K.*

Proof. Let $p \in K$. By Lemma 2 choose j, k with $p \in W_j \cap V_k$; then $t_0, s_0 \ge 0$. As $Z_{jk} < 0$, $t_0 < T_j$ or $s_0 < T_k$, so by Lemma 1 p is in the j-th cylinder of the first family or the k-th of the second. □

The hypothesis is sufficient, not necessary: a point of $W_j \cap V_k$ with $Z > 0$ there can still be covered by a neighbouring sector. The preprint's radial argument (lines 47–124 of `coverage.tex`) is one way to bound all Z_{jk} at once; Theorem 3 replaces it by a finite exact computation and allows a separate cutoff per sector.

Area. $|P_j| = HT_j^2/n$ and the normal of the plane $x = 0$ has inner product 1 with v_j, so the perpendicular base has area $|P_j|/|v_j|$, $|v_j|^2 = 1 + \varepsilon^2(\alpha_j^2 + 2\beta_j^2)$; likewise for Q_j, w_j. Hence

$$\frac{\sum_i |B_i|}{A_{\min}(K)} = \sum_{j=0}^{n-1} \frac{2T_j^2}{n\sqrt{1 + \varepsilon^2(\alpha_j^2 + 2\beta_j^2)}}. \tag{1}$$

At $\varepsilon = 0$, $T_j = \frac{1}{2}$ this is exactly $\frac{1}{2}$ (Bang's cover).

4. The certificate

Theorem 4. *Let $\varepsilon = 2/5$, $n = 320$, and $T_j = \frac{1}{2} + \varepsilon^2(\max(d(q_j), d(q_{j+1})) + E_j)$ with the rational margins $E_j \in [2.39 \cdot 10^{-5}, 2.67 \cdot 10^{-3}]$ listed in `scripts/independent/params_T.txt` (md5*

0f4291bd4ee5cd4686db5d688b8d0f1e). The resulting 640 *cylinders, with compact nondegenerate triangular perpendicular bases, cover* K*, and*

$$0.4999773398010235\,4 \le \frac{\sum_i |B_i|}{A_{\min}(K)} \le 0.4999773398010235\,5 < \frac{1}{2}.$$

Consequently also $\sum_i |B_i|/|\pi_{u_i^\perp} K| < \frac{1}{2}$ *(numerically* ≈ 0.4511*).*

Computational proof. Coverage. For all 102,400 pairs (j,k), Z_{jk} was computed exactly over $\mathbb{Q}$ (SageMath, PPL backend). The maximum is $Z^* = -2.39999977 \cdot 10^{-8} < 0$, attained at $(78, 84)$ and its three images under the symmetries of the construction; apply Theorem 3. Independently, the margins were certified by exact nonnegative dual vectors y with $y^\top A_{x,y,t} = 0$, $y^\top a_z = 1$, giving $Z_{jk} \le \varepsilon^2 U_{jk}$ with $\max U_{jk} = -1.5 \cdot 10^{-7}$, which equals Z^*/ε^2. *Area.* (1) evaluated in 256-bit interval arithmetic with exact rational inputs gives the stated enclosure; a second evaluation with rational upper bounds $r_j \ge |v_j|^{-1}$ (checked by $r_j^2 |v_j|^2 \ge 1$) gives 0.4999773398. *Directionwise.* Each denominator is $\ge A_{\min}(K)$, as in [4, eq. `cost-comparison`]. □

Remark 5 (sanity checks). (i) 1,102,405 points of K with exact rational coordinates (uniform; $|t - \frac{1}{2}| \le \frac{1}{20}$; on faces; on edges and within 10^{-3} of vertices; the 102,400 LP maximizers) were each verified to lie in one of the 640 cylinders by Lemma 1 in exact arithmetic; none failed. The same test with all $T_j = \frac{1}{2}$ finds uncovered points (959 of 40,005), so it is not vacuous. (ii) With the same axes, cutoffs $T_j = \frac{1}{2} + \varepsilon^2(d_j + \max_i E_i)$ (a uniform margin) give area $0.50056 > \frac{1}{2}$, so per-sector cutoffs are needed at this ε. (iii) A numerical minimization over $2 \cdot 10^5$ directions reproduces $A_{\min}(K) = \sqrt{2}$.

5. Where the constant was lost

The preprint bounds all Z_{jk} through one inequality, $2\eta - (2 + 2\eta)\varepsilon - \varepsilon^2/4 > 0$ (`coverage.tex`, eq. `uniform-coverage-gap`). The term $(2+2\eta)\varepsilon$ comes from the rough-angle estimate $|y - q/2|, |x - p/2| \le \varepsilon$ and $|x^2 - p^2/4| \le \frac{3}{2}\varepsilon$, valid uniformly but first-order in ε, while the gain is only 2η; with $\eta = 10^{-3}$ this forces $\varepsilon \le 1/2000$ and $n = \lceil 2/\varepsilon^2 \rceil = 8 \cdot 10^6$ sectors per family. Area saving $\frac{13}{6000}\varepsilon^2$ is then about $5 \cdot 10^{-10}$. The exact formulation does not pay these first-order terms, and the margin actually required varies by two orders of magnitude across sectors (the largest E_j sit at the two ends $|q| \approx 1$, the smallest near $q = 0$), so a per-sector margin pays only where it is needed.

	ε	cylinders	$\sum \lvert B_i\rvert / A_{\min}$
Preprint, Thm. 1.1	1/2000	16,000,000	$\frac{1}{2} - \frac{13}{6000}\varepsilon^2 + O(\varepsilon^4) \approx \frac{1}{2} - 5.4 \cdot 10^{-10}$
This note	2/5	640	$\le \frac{1}{2} - 2.266 \cdot 10^{-5}$

The count improves by a factor 25,000 over the displayed parameters, and by a factor 1,562.5 over the 10^6 cylinders obtainable inside the preprint's certificate [5]. We do not claim 640 is minimal for this family; with $n = 128$ sectors the same procedure (heuristic margins, not certified) gave area $0.50045 > \frac{1}{2}$.

What is not claimed. No new idea beyond the preprint's construction: the selection lemma, axes, triangles and area formula are OpenAI's; only the cutoffs, the parameters and the certificate are new. The certificate is a computer proof (exact rational LP via PPL, plus an independent exact-dual check); it has not been formalized in Lean.

Authorship. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

References

[1] T. Bang, A solution of the "plank problem", Proc. Amer. Math. Soc. 2 (1951), 990–993.
[2] K. Bezdek, Tarski's plank problem revisited, arXiv:0903.4637 (2009), Problem 3.1.

[3] K. Bezdek and A. E. Litvak, Covering convex bodies by cylinders and lattice points by flats, J. Geom. Anal. 19 (2009), 233–243; arXiv:0903.4842.

[4] OpenAI, Finite angular cylinder covers below the half-area bound (September 27, 2026), `https://github.com/openai/math/tree/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/Finite-angular-cylinder-covers-below-the-half-area-bound-September-27-2026`.

[5] supsup/new_math_review, *Covering a tetrahedron with a million thin cylinders* (family-100 card, `showcase.md`; see also `possible_further_findings.md`, item 1), commit `cd6549b2`, 2026-10-08, `https://github.com/supsup/new_math_review`.